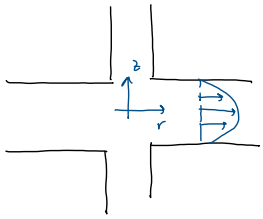
7.10

Monday, 6 November 2023 5:47 PM

Radial flow b/w parallel plates



$P(r)$

$v_r(r,z)$

+ Continuity: $v_r = \frac{f(z)}{r}$ (arbitrary function of z (which we don't know!))

+ Navier stokes:

r-component: $\rho\left(v_r \frac{\partial v_r}{\partial r}\right) = -\frac{\partial P}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2}$

Although v_r is changing w/ r (as you go further away to the periphery)

We can assume this process is slow enough that only variation in v_r in z-direction dominates

This is the "slow" or developed flow assumption

Now we have:

$$\frac{\partial P}{\partial r} = \mu \frac{\partial^2 v_r}{\partial z^2}$$

We have said $\frac{\partial v_r}{\partial r} \approx 0$ but that does not mean $v_r \neq f(r)$

Cannot solve this expression as of yet since v_r is $f(r,z)$ so cannot take integral of both sides!

However, substitute $v_r = \frac{1}{r} f(z)$

$$\frac{dP}{dr} = \mu \frac{1}{r} \frac{d^2 f}{dz^2}$$

$$r \frac{dP}{dr} = \mu \frac{d^2 f}{dz^2} = \lambda$$

Now separation of variables is possible! (since f is only function of z & P only of r)

λ: Common parameter

$r \frac{dP}{dr} = \lambda$

BC's ✓

↓

solve for λ here

↓

Which we can plug in here at the end

Check OH of Exam 2 review session!

$\frac{d^2 f}{dz^2} = \frac{\lambda}{\mu}$

BC's ✓

↓

Solve for f

↓

get v_r